

Twistor Geometry for Hyperkähler Metrics on Complex Adjoint Orbits

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Abstract. We study the hyperkähler geometry of complex adjoint orbits from the point of view of twistor theory. We introduce, for complex semisimple adjoint orbits, the associated spectral curve and construct the twistor space as a union of certain regular adjoint orbits; we also exhibit the family of twistor lines. Furthermore, we show how our methods may be applied for describing hyperkähler metrics associated to more general spectral curves. In particular, we give an algebraic characterisation of the twistor lines.

Key words: adjoint orbits, hyperkähler metrics, twistor spaces

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1. Introduction

The study of hyperkähler metrics on complex adjoint orbits has been undertaken in at least two different ways. On the one hand, there is the twistorial approach of Burns [4]. In fact, Burns was originally concerned with generalising Calabi's hyperkähler metrics on the cotangent bundle of projective space so as to describe hyperkähler metrics on cotangent bundles of flag manifolds, but his method serves as well to investigate complex adjoint orbits, since these arise as generic fibres of Burns' twistor spaces. More precisely, Burns studies hyperkähler metrics on $T^*(G^c/B_+)$, depending on the choice of a regular element $z \in \mathfrak{t}$. The twistor space $Z_{B,z}$ for such a metric is constructed algebraically, essentially as a resolution of singularities $q_B : Z_{B,z} \rightarrow \overline{X}$; here $\overline{X} \subseteq \mathfrak{g}^c \otimes \mathcal{O}(2)$ is obtained by glueing two copies of the Zariski closure of

$$\{(\rho, \zeta) \in \mathfrak{g}^c \times \mathbb{C}^* / \rho = \text{Ad}(g)(\zeta z) \text{ for some } g \in G^c\}$$

by the map $\mathfrak{g}^c \times \mathbb{C}^* \rightarrow \mathfrak{g}^c \times \mathbb{C}^*$, $(\rho, \zeta) \mapsto (\zeta^{-2}\rho, \zeta^{-1})$. There is, however, a shortcoming in his procedure arising from the difficulty in exhibiting the twistor lines. In order to make his methods work, Burns has to rely on an existence theorem of relative deformation theory, so as to guarantee the existence of twistor lines.

On the other hand, there is Kronheimer's method [16] (to be described in some detail in Section 2), in which the adjoint orbit is identified with a certain moduli space of instantons; the hyperkähler property then follows from the fact

that the latter may be constructed as a hyperkähler quotient (or rather an infinite-dimensional version of it). However, by going through this process to describe a metric on the adjoint orbit, one has to rely on existence results for solving differential equations.

We propose a different procedure for studying hyperkähler metrics on complex adjoint orbits, which however owes to both Burns' and Kronheimer's approaches. We construct a twistor space X_S , which depends on the choice of a spectral curve S , and are able to give a characterisation of twistor lines. For particular choices of the curve S , we are in the case of complex regular semisimple orbits and we may, with the help of Kronheimer's work, describe a *family* of twistor lines on X_S , whereas Burns relies on Kodaira's relative deformation theory to guarantee the existence of such a family. These results are explained in Sections 3 and 4. Finally, in Section 5 we discuss the construction of the metric by means of the identification of a fibre of the twistor space with the parameter space of twistor lines.

Let us fix some notations and conventions. Let G denote a compact semisimple Lie group, G^c its complexification, and let $\mathfrak{g}$ and $\mathfrak{g}^c$ denote their Lie algebras, respectively. Also fix a maximal torus T in G and let $\mathfrak{t}$ be its Lie algebra. We recall that the Killing form on $\mathfrak{g}^c$ is non-degenerate and G^c -invariant and hence allows us to identify adjoint and coadjoint complex orbits; we shall use this identification at our convenience, without further notice. We denote the G^c -adjoint orbit of $x \in \mathfrak{g}^c$ by $\mathcal{O}_x$. Finally, recall that $\mathfrak{g}$ is the fixed-point set of the *Cartan involution* (the usual *real structure* on $\mathfrak{g}^c$); we write this involution simply as $X \mapsto -X^*$. This map has its usual meaning when $\mathfrak{g}$ is a Lie subalgebra of $\mathfrak{su}_k$ so that $\mathfrak{g}^c$ is a Lie subalgebra of $\mathfrak{sl}(k, \mathbb{C})$.

2. The Kronheimer–Biquard Construction

We first recall the construction of complex semisimple adjoint orbits as hyperkähler moduli spaces. This may be done either by Kronheimer's approach in [16] using $SU(2)$ -invariant G -instantons on $\mathbb{R}^4 - 0$, or by a variation of his methods due to Biquard in [3]. We shall opt for the latter. Biquard's construction can, in fact, be generalised to give hyperkähler metrics on any complex adjoint orbit of G^c (see also [15]), but in this and the next section we shall only deal with the semisimple case.

Following Biquard, we consider connections on $(-\infty, 0] \times S^1 \times S^1 \times S^1$ which are invariant with respect to rotations of each of the circle factors, that is to say, which are of the type $A = d + B_0(s)ds + \sum_{i=1}^3 B_i(s)d\theta_i$, for smooth maps $B_i : (-\infty, 0] \rightarrow \mathfrak{g}$, $i = 0, 1, 2, 3$. We represent such a connection by the quadruple (B_0, B_1, B_2, B_3) . The anti-self-duality condition on the curvature of A , $*F(A) = -F(A)$, is equivalent to the three equations

$$\frac{dB_i}{ds} = -[B_0, B_i] - [B_j, B_k], \quad (1)$$

where (i, j, k) ranges over the cyclic permutations of $(1, 2, 3)$. We may then consider the space of solutions of (1) modulo the action of smooth gauge transformations $g : (-\infty, 0] \rightarrow G$ such that $g(0) = 1$. Here g acts by $(B_0, B_1, B_2, B_3) \mapsto (-dg/ds g^{-1} + \text{Ad}(g)B_0, \text{Ad}(g)B_1, \text{Ad}(g)B_2, \text{Ad}(g)B_3)$. We denote a point in this moduli space by $[(B_0, B_1, B_2, B_3)]$. There is a gauge transformation which makes $B_0 = 0$, which is unique since we require $g(0) = 1$. So we may identify the space above with the space of solutions to the equations

$$\frac{dB_1}{ds} = -[B_2, B_3], \quad \frac{dB_2}{ds} = -[B_3, B_1], \quad \frac{dB_3}{ds} = -[B_1, B_2]. \quad (2)$$

Now fix a triple $(\tau_1, \tau_2, \tau_3) \in \mathfrak{t}^3$ which is regular, in the sense that its centraliser in $\mathfrak{g}$ is just $\mathfrak{t}$. We are concerned here with the space $M(\tau_1, \tau_2, \tau_3)$ consisting of solutions to Equations (2) such that

$$B_i(s) \rightarrow \text{Ad}(g_0)(\tau_i) \text{ as } s \rightarrow -\infty, \text{ for some } g_0 \in G, \quad i = 1, 2, 3. \quad (3)$$

This is equivalent to considering solutions of (1) which approach the “model solution” $\Delta = d + \sum_{i=1}^3 \tau_i d\theta_i$ at $-\infty$, modulo gauge transformations.

Let us also recall that Biquard considers the space of connections $\mathcal{A}_\epsilon = \{\Delta + a, a \in \Omega_{\Delta, \epsilon}^1\}$, where $\Omega_{\Delta, \epsilon}^1 = \{a, |a| = O(s^{-(1+\epsilon)}), |\Delta a| = O(s^{-(2+\epsilon)})\}$, and the gauge group $\mathcal{G}_\epsilon = \{g : (-\infty, 0] \rightarrow G, g(0) = 1, (\Delta g)g^{-1} \in \Omega_{\Delta, \epsilon}^1\}$. Biquard then shows that there is an $\epsilon > 0$ such that

$$M(\tau_1, \tau_2, \tau_3) = \{[(B_0, B_1, B_2, B_3)] \in \mathcal{A}_\epsilon / \mathcal{G}_\epsilon, [(B_0, B_1, B_2, B_3)] \text{ satisfies (1)}\}$$

and defines a manifold structure on this space.

We recall that a hyperkähler structure on a manifold M consists of a Riemannian metric g and three almost-complex structures I, J, K satisfying the quaternionic relations, such that (M, g) is Kähler with respect to each of them. The hyperkähler metric on $M(\tau_1, \tau_2, \tau_3)$ is defined as follows. One considers first a metric on $\mathcal{A}_\epsilon$, which is an affine space modelled on $\Omega_{\Delta, \epsilon}^1$, by defining for $a, b \in \Omega_{\Delta, \epsilon}^1$,

$$g(a, b) = \int_{-\infty}^0 \langle a(s), b(s) \rangle ds.$$

Here, $\langle \rangle$ denotes an invariant inner product on $\mathfrak{g}$, which for definiteness we take to be $-K$, where K is the Killing form of $\mathfrak{g}$. The almost-complex structures I, J, K are defined by $I(a_0, a_1, a_2, a_3) = (-a_1, a_0, -a_3, a_2)$, $J(a_0, a_1, a_2, a_3) = (-a_2, a_3, a_0, -a_1)$, $K(a_0, a_1, a_2, a_3) = (-a_3, -a_2, a_1, a_0)$, and the symplectic forms by $\omega_1(a, b) = g(Ia, b)$, $\omega_2(a, b) = g(Ja, b)$, $\omega_3(a, b) = g(Ka, b)$. Now the tangent space to $M(\tau_1, \tau_2, \tau_3)$ at a point $[A] = [(B_0, B_1, B_2, B_3)]$ may be

identified with the space of $a \in \Omega_{\Delta, \epsilon}^1$ such that $d_A^* a = 0$, $d_A^+ a = 0$, and these conditions may be written down as

$$\left. \begin{aligned} \frac{da_0}{ds} + [B_0, a_0] + [B_1, a_1] + [B_2, a_2] + [B_3, a_3] &= 0, \\ \frac{da_1}{ds} + [B_0, a_1] - [B_1, a_0] + [B_2, a_3] - [B_3, a_2] &= 0, \\ \frac{da_2}{ds} + [B_0, a_2] - [B_1, a_3] - [B_2, a_0] + [B_3, a_1] &= 0, \\ \frac{da_3}{ds} + [B_0, a_3] + [B_1, a_2] - [B_2, a_1] - [B_3, a_0] &= 0. \end{aligned} \right\}. \quad (4)$$

One then verifies that the almost-hyperkähler structure on $\mathcal{A}_\epsilon$ descends to a hyperkähler structure on $M(\tau_1, \tau_2, \tau_3)$. The integrability of the almost-complex structures follows from the closedness of the symplectic forms [6]. In fact we may formally describe $M(\tau_1, \tau_2, \tau_3)$ as a hyperkähler quotient; this is similar to the construction of moduli spaces for Higgs bundles in [6] and for magnetic monopoles in [2].

It turns out that, when $\tau_2 + i\tau_3$ is a regular element of $\mathfrak{g}^c$, $M(\tau_1, \tau_2, \tau_3)$ equipped with the complex structure I can be identified as a complex manifold with the complex adjoint orbit $\mathcal{O}_{\tau_2 + i\tau_3}$, an explicit biholomorphism being provided by the map $M(\tau_1, \tau_2, \tau_3) \rightarrow \mathcal{O}_{\tau_2 + i\tau_3}$ defined by $[(B_0, B_1, B_2, B_3)] \mapsto B_2(0) + iB_3(0)$. Furthermore, $M(\tau_1, \tau_2, \tau_3)$ is a holomorphic-symplectic manifold with the complex symplectic form $\omega^c = \omega_2 + i\omega_3$ and the identification above preserves the holomorphic symplectic structures, that is to say, ω^c is the pull-back of the Kostant–Kirillov form on $\mathcal{O}_{\tau_2 + i\tau_3}$.

The adjoint action of the compact group G induces an action on $M(\tau_1, \tau_2, \tau_3)$ preserving its hyperkähler structure.

LEMMA 1.* *The moment map $\mu_i : M(\tau_1, \tau_2, \tau_3) \rightarrow \mathfrak{g}$ for the action of G on the symplectic manifold $(M(\tau_1, \tau_2, \tau_3), \omega_i)$ is given by $\mu_i = B_i(0)$.*

Proof. We shall use the description $M(\tau_1, \tau_2, \tau_3) = \mathcal{A}_\epsilon / \mathcal{G}_\epsilon$ as above. The action of G lifts to an action on $\mathcal{A}_\epsilon$, and preserves the symplectic forms ω_i there. In order to compute the moment map, consider $\varrho \in \mathfrak{g}$ and let $\{e^{t\varrho}\}$ be the associated 1-parameter subgroup of G . The corresponding vector field on $\mathcal{A}_\epsilon$ is given by

$$X^\varrho(B_0, B_1, B_2, B_3) = ([\varrho, B_0], [\varrho, B_1], [\varrho, B_2], [\varrho, B_3]).$$

Let $Y = (b_0, b_1, b_2, b_3)$ be a tangent vector to $M(\tau_1, \tau_2, \tau_3)$ at $[(B_0, B_1, B_2, B_3)]$; we easily compute

$$\omega_1(X^\varrho, Y) = \int_{-\infty}^0 \langle \varrho, -[B_1, b_0] + [B_0, b_1] - [B_3, b_2] + [B_2, b_3] \rangle ds.$$

In view of (4), we may rewrite this as

$$- \int_{-\infty}^0 \frac{d}{ds} \langle \varrho, b_1 \rangle ds = -\langle \varrho, b_1(0) \rangle,$$

* This is essentially stated in [16].

since $\lim_{s \rightarrow -\infty} b_1(s) = 0$. We have thus shown that $\omega_1(X^\varrho, Y) = K(\varrho, b_1(0))$. Now the map $\mu_1^\varrho : [(B_0, B_1, B_2, B_3)] \mapsto K(\varrho, B_1(0))$ has differential given by $d\mu_1^\varrho(Y) = \omega_1(X^\varrho, Y)$, so that $\iota(X^\varrho)\omega_1 = d\mu_1^\varrho$. We conclude that the (G -equivariant) map $\mu_1 : M(\tau_1, \tau_2, \tau_3) \rightarrow \mathfrak{g}^*$ defined by $\mu_1([(B_0, B_1, B_2, B_3)])(\varrho) = K(\varrho, B_1(0))$ is the required moment map. Under the isomorphism $\mathfrak{g} \cong \mathfrak{g}^*$ provided by the Killing form K , we have $\mu_1([(B_0, B_1, B_2, B_3)]) = B_1(0)$. We deal with μ_2 and μ_3 similarly. $\square$

3. Spectral Curve and Twistor Space

Let us first recall some facts about twistor spaces associated to hyperkähler manifolds, the proofs of which may be found in [9]. Given a hyperkähler manifold M , its *twistor space* Z is the product manifold $M \times S^2$ equipped with the complex structure $\underline{I}$, where $\underline{I} = (aI + bJ + cK, I_0)$ at the point $(m, (a, b, c))$ ($m \in M$, $(a, b, c) \in S^2$); here I_0 is the usual complex structure on S^2 , which we identify with $\mathbb{P}^1$. An application of the Newlander–Nirenberg theorem shows that $\underline{I}$ is integrable, so that Z is a complex manifold. Furthermore, the projection $p : Z \rightarrow \mathbb{P}^1$ is holomorphic, and the antipodal map of S^2 induces a real structure on Z . The *twistor lines* are the copies $(m, \mathbb{P}^1)$ of the projective line; the normal bundle of a twistor line is isomorphic to $\mathcal{O}(1) \oplus \cdots \oplus \mathcal{O}(1)$. One obtains a holomorphic symplectic form along the fibres of $p : Z \rightarrow \mathbb{P}^1$ by $\omega = (\omega_2 + i\omega_3) + 2\zeta\omega_1 - \zeta^2(\omega_2 - i\omega_3)$. More precisely, this defines $\omega \in H^0(Z; \wedge^2 T_F^*(2))$, where T_F denotes the (holomorphic) tangent bundle along the fibres of $p : Z \rightarrow \mathbb{P}^1$.

We consider next the twistor space for Kronheimer's hyperkähler metric on $M(\tau_1, \tau_2, \tau_3)$. Letting I, J, K denote the complex structures as described in the previous section, it will be convenient to consider the ordered triple $\{J, -I, K\}$. Observe that this triple satisfies the canonical quaternionic relations. Carrying out the construction described above, we obtain the twistor space for the hyperkähler manifold $M(\tau_1, \tau_2, \tau_3)$, which we shall denote by $Z_{(\tau_1, \tau_2, \tau_3)}$. A point in $M(\tau_1, \tau_2, \tau_3)$ may be written as a triple (B_1, B_2, B_3) satisfying Equations (2) and conditions (3); it is also convenient to change notation slightly and write $T_1 = -B_3$, $T_2 = -B_1$, $T_3 = -B_2$, so that we represent a point in $M(\tau_1, \tau_2, \tau_3)$ by a triple (T_1, T_2, T_3) of maps $T_i : (-\infty, 0] \rightarrow \mathfrak{g}$ satisfying *Nahm's equations*

$$\frac{dT_1}{ds} = [T_2, T_3], \quad \frac{dT_2}{ds} = [T_3, T_1], \quad \frac{dT_3}{ds} = [T_1, T_2]$$

and such that as $s \rightarrow -\infty$,

$$T_1(s) \rightarrow -\text{Ad}(g_0)(\tau_3), T_2(s) \rightarrow -\text{Ad}(g_0)(\tau_1), T_3(s) \rightarrow -\text{Ad}(g_0)(\tau_2),$$

for some $g_0 \in G$. We denote a point in $Z_{(\tau_1, \tau_2, \tau_3)}$ by $[(T_1, T_2, T_3), \zeta]$, for $(T_1, T_2, T_3) \in M(\tau_1, \tau_2, \tau_3)$, $\zeta \in \mathbb{P}^1$.

Recall that we can rewrite Nahm's equations in *Lax form* by letting $A(s, \zeta) = T_1 + iT_2 + \zeta(-2iT_3) + \zeta^2(T_1 - iT_2)$, $A_+(s, \zeta) = -iT_3 + \zeta(T_1 - iT_2)$, where ζ is a complex parameter; Nahm's equations are equivalent to $(dA/ds) = [A, A_+]$. Notice that if we write $A(s, \zeta) = A_0 + A_1\zeta + A_2\zeta^2$, where $A_i : (-\infty, 0] \rightarrow \mathfrak{g}^c$, $i = 1, 2, 3$, then $A_+ = (1/2)A_1 + A_2\zeta$. Also observe that the conditions $T_i^* = -T_i$ are equivalent to $A_0^* = -A_2$, $A_1^* = A_1$. These are referred to as the *reality conditions* on A .

Let us now recall how to obtain the spectral curve associated to $A(s, \zeta)$ [5]. We shall assume that $\mathfrak{g}^c \subset \mathfrak{gl}(k, \mathbb{C})$. The isospectral property of Lax-type equations says that the characteristic polynomial of $A(s, \zeta)$ is a "constant of motion", that is, does not depend on s . The curve given by the equation $\det(\eta + A) = 0$ in the (η, ζ) -plane may be compactified to a curve in $T\mathbb{P}^1$ as follows. The expression $\det(\eta + A)$ has a global interpretation as a section $\psi \in H^0(T\mathbb{P}^1; \mathcal{O}(2k))$; this is described by $\psi_0(\eta, \zeta) = \det(\eta + A)$ on $\tilde{U}_0$ and $\psi_\infty(\tilde{\eta}, \tilde{\zeta}) = \det(\tilde{\eta} + \tilde{A})$ on $\tilde{U}_\infty$; here $\tilde{A} = A_0\tilde{\zeta}^2 + A_1\tilde{\zeta} + A_2$ and $\{\tilde{U}_0, \tilde{U}_\infty\}$ is the open covering of $T\mathbb{P}^1$ given by $\tilde{U}_i = \pi^{-1}(U_i)$ ($i = 0, \infty$), where U_0, U_∞ is the usual open covering of $\mathbb{P}^1$. One readily verifies that $\psi_0(\eta, \zeta) = \zeta^{2k}\psi_\infty(\eta/\zeta^2, \zeta^{-1})$ on $\tilde{U}_0 \cap \tilde{U}_\infty$, so that ψ_0, ψ_∞ define a section of $\pi^*\mathcal{O}(2k) \rightarrow \mathcal{O}(2)$, where $\pi : \mathcal{O}(2) \rightarrow \mathbb{P}^1$ is the projection. Here we are using the same notation for the bundles and their corresponding total spaces. Recalling that we may identify $\mathcal{O}(2)$ with $T\mathbb{P}^1$, and denoting $\pi^*\mathcal{O}(2k)$ simply by $\mathcal{O}(2k)$, we get $\psi \in H^0(T\mathbb{P}^1; \mathcal{O}(2k))$. The *spectral curve* S associated to A is the divisor of the section ψ . We shall usually refer to it simply by writing down the affine part $S : \det(\eta + A) = 0$.

We now show that all Nahm solutions which belong to $M(\tau_1, \tau_2, \tau_3)$ give rise to the same spectral curve, so that we may speak of *the spectral curve of the adjoint orbit* $\mathcal{O}_{\tau_2+i\tau_3}$. In fact given (T_1, T_2, T_3) belonging to $M(\tau_1, \tau_2, \tau_3)$, consider the associated $A(s, \zeta)$. Because of the isospectral property of $A(s, \zeta)$, $\det(\eta + A(s, \zeta)) = \lim_{s \rightarrow -\infty} \det(\eta + A(s, \zeta))$; using the boundary conditions, we have $\lim_{s \rightarrow -\infty} A(s, \zeta) = -\text{Ad}(g_0)((\tau_3 + i\tau_1) + \zeta(-2i\tau_2) + \zeta^2(\tau_3 - i\tau_1))$ so that $\lim_{s \rightarrow -\infty} \det(\eta + A(s, \zeta)) = \det(\eta - ((\tau_3 + i\tau_1) + \zeta(-2i\tau_2) + \zeta^2(\tau_3 - i\tau_1)))$, thus proving our claim.

We also recall from [9] that if a group G acts on a hyperkähler manifold by isometries, then the induced action of G on Z is by holomorphic maps. For each $\zeta \in \mathbb{P}^1$, G acts on the complex-symplectic manifold $(M(\tau_1, \tau_2, \tau_3), \underline{I}(\zeta))$ by holomorphic maps preserving its holomorphic symplectic form. If the action has moment maps μ_1, μ_2, μ_3 corresponding to the Kähler forms $\omega_1, \omega_2, \omega_3$, respectively, the twisted symplectic form ω described previously gives rise to a twisted moment map globally defined on Z ; more precisely, this is the holomorphic map $\mu : Z \rightarrow \mathfrak{g}^c \otimes \mathcal{O}(2)$ given by $\mu = (\mu_2 + i\mu_3) + 2\zeta\mu_1 - \zeta^2(\mu_2 - i\mu_3)$. Let us apply this construction to the G -action on $M(\tau_1, \tau_2, \tau_3)$.

PROPOSITION 2. *The holomorphic moment map $\mu : Z_{(\tau_1, \tau_2, \tau_3)} \rightarrow \mathfrak{g}^c \otimes \mathcal{O}(2)$, associated to the action of G on $M(\tau_1, \tau_2, \tau_3)$, is given by $\mu([(T_1, T_2, T_3), \zeta]) = -i A(0, \zeta)$.*

Here $A(0, \zeta)$ is the evaluation at $s = 0$ of the solution $A(s, \zeta)$, corresponding to (T_1, T_2, T_3) , of the Lax-type equation $dA/ds = [A, A_+]$.

Proof. This is now straightforward in view of Lemma 1 and the comments above. Just observe that as the ordered triple of complex structures out of which the twistor space is built is $\{J, -I, K\}$, the respective symplectic forms are $\omega_2, -\omega_1, \omega_3$ (we are again using the notation of the previous section). Therefore, the expression for the holomorphic symplectic form along the fibres of $p : Z_{(\tau_1, \tau_2, \tau_3)} \rightarrow \mathbb{P}^1$ becomes $\omega = -i[(-\omega_3 - i\omega_1) + 2i\zeta\omega_2 + \zeta^2(-\omega_3 + i\omega_1)]$, so that the associated complex moment map is given by $\mu = -i[(-\mu_3 - i\mu_1) + 2i\zeta\mu_2 + \zeta^2(-\mu_3 + i\mu_1)]$. Thus according to Lemma 1, we get $i\mu = (T_1(0) + iT_2(0)) - 2i\zeta T_3(0) + \zeta^2(T_1(0) - iT_2(0))$ and the last expression is just $A(0, \zeta)$. $\square$

In order to proceed in a more algebraic fashion, we shall consider instead of $Z_{(\tau_1, \tau_2, \tau_3)}$ a slightly smaller twistor space X , which will account for the hyperkähler metric on an open dense submanifold of $M(\tau_1, \tau_2, \tau_3)$. Observe that the adjoint action induces an action of G^c on $\mathfrak{g}^c \otimes \mathcal{O}(2)$ and so it makes sense to talk about regular elements in $\mathfrak{g}^c \otimes \mathcal{O}(2)$. We define X as the regular part of the image of $Z_{(\tau_1, \tau_2, \tau_3)}$ by the map $q = i\mu$. Alternatively, we may describe X as the manifold generated by a regular section in $H^0(\mathbb{P}^1, \mathfrak{g}^c \otimes \mathcal{O}(2))$ having spectral curve S . From now on we take the point of view of studying the twistor geometry of X . Although the latter description makes it possible to describe X with no reference to $Z_{(\tau_1, \tau_2, \tau_3)}$, it is the relationship between these two twistor spaces which allows us to describe a family of twistor lines on X . In fact consider the image of a twistor line in $Z_{(\tau_1, \tau_2, \tau_3)}$ by the map q ; we shall call such a section of $\mathfrak{g}^c \otimes \mathcal{O}(2)$ a *Kronheimer line*.

THEOREM 3. *Regular Kronheimer lines are twistor lines in X .*

Proof. We must show that each regular Kronheimer line Φ is a real section of $\mathfrak{g}^c \otimes \mathcal{O}(2)$ such that the normal bundle of Φ in X is isomorphic to $\mathcal{O}(1) \oplus \cdots \oplus \mathcal{O}(1)$. Reality follows immediately, and so all we have to check is the normal bundle condition. Let us denote $Z_{(\tau_1, \tau_2, \tau_3)}$ simply by Z , and the fibre over ζ of $Z \rightarrow \mathbb{P}^1$ by Z_ζ . First let us consider a point $p \in Z$ such that $q(p)$ is regular. Since G^c acts transitively on $\mathcal{O}_{q(p)}$ (by definition!), the tangent space $T\mathcal{O}_{q(p)}$ is spanned by the vectors $X_{\mathcal{O}}^\rho(q(p))$ for $\rho \in \mathfrak{g}^c$; here, $X_{\mathcal{O}}^\rho$ denotes the vector field on $\mathcal{O}_{q(p)}$ generated by ρ . Note that if $\rho = \rho_1 + i\rho_2$, where $\rho_1, \rho_2 \in \mathfrak{g}$, then $X_{\mathcal{O}}^\rho = X_{\mathcal{O}}^{\rho_1} + iX_{\mathcal{O}}^{\rho_2}$. Now consider the map $q|_{Z_\zeta} : Z_\zeta \rightarrow \mathfrak{g}^c$. Because of the G -invariance of q , we have

$$dq_p(X_{Z_\zeta}^{\rho_j}(p)) = X_{\mathcal{O}}^{\rho_j}(q(p)), \quad \text{for } j = 1, 2.$$

Now since q is holomorphic,

$$dq_p(I_\zeta X_{Z_\zeta}^{\rho_2}(p)) = idq_p(X_{Z_\zeta}^{\rho_2}(p)) = iX_{\mathcal{O}}^{\rho_2}(q(p)).$$

Thus, we have the identity

$$dq_p(X_{Z_\zeta}^{\rho_1}(p) + I_\zeta X_{Z_\zeta}^{\rho_2}(p)) = X_{\mathcal{O}}^\rho(q(p)),$$

from which we see that the differential $dq_p : T(Z_\zeta)_p \rightarrow T\mathcal{O}_{q(p)}$ is surjective (since the vectors $X_{\mathcal{O}}^\rho(q(p))$ span $T\mathcal{O}_{q(p)}$, as we remarked above). However, dq_p is a linear map between spaces of the same dimension, namely $\dim(\mathfrak{g}^c) - \text{rank}(\mathfrak{g}^c)$; therefore, dq_p is one to one.

Now if Ψ is a section of $Z \rightarrow \mathbb{P}^1$ such that $\Phi = q \circ \Psi$ is regular, then we have for each $\zeta \in \mathbb{P}^1$ an isomorphism

$$dq_{\Psi(\zeta)} : T(Z_\zeta)_{\Psi(\zeta)} \rightarrow T\mathcal{O}_{\Phi(\zeta)}.$$

Hence dq along the line Ψ allows us to holomorphically identify the tangent bundle of Ψ along the fibres of $Z \rightarrow \mathbb{P}^1$ and the tangent bundle of Φ along the fibres of $X \rightarrow \mathbb{P}^1$; or, which is the same, we may identify the normal bundle of Ψ in Z with the normal bundle of Φ in X . However, if Ψ is a twistor line in Z , it has normal bundle $\mathcal{O}(1) \oplus \cdots \oplus \mathcal{O}(1)$, and therefore so does Φ . $\square$

Remark. We have seen that we may obtain twistor lines in X by evaluating at $s = 0$ the solutions to the Lax-type equation having a fixed spectral curve. On the other hand, as is well known [5], these solutions may be obtained in an algebraic fashion from line bundles on the spectral curve. We shall exploit this relationship further in a forthcoming paper.

4. An Algebraic Characterisation of Regular Twistor Lines

We shall now consider the question of extending the previous discussion so as to allow more general spectral curves. In this way we shall be dealing not only with Kronheimer's metrics but also with more general hyperkähler metrics which admit a similar twistor description. We thus consider, as before, the complex manifold $X_S \subseteq \mathfrak{g}^c \otimes \mathcal{O}(2)$ associated to a spectral curve S . More precisely, this is defined as the orbit under the adjoint action of G^c of a regular section in $H^0(\mathbb{P}^1, \mathfrak{g}^c \otimes \mathcal{O}(2))$ with spectral curve S . Note that the fibre over $\zeta \in \mathbb{P}^1$ of the projection $X_S \rightarrow \mathbb{P}^1$ may be identified with the (regular) adjoint orbit $\mathcal{O}_{\Phi(\zeta)}$. These manifolds X_S have also been considered in [1]; however, our main concern here is a characterisation of twistor lines in X_S , which is not considered in that paper.

THEOREM 4. *Let Φ be a real section of $X_S \rightarrow \mathbb{P}^1$. Then Φ is a twistor line if and only if its bundle of centralisers fills the Lie algebra $\mathfrak{g}^c$.*

What we mean in the last statement is that the fibres of the bundle of centralisers, viewed as a subbundle of the trivial bundle over $\mathbb{P}^1$ with fibre $\mathfrak{g}^c$, span $\mathfrak{g}^c$ as a vector space. The remainder of this section is devoted to proving this result.

Denote by X_ζ the fibre over ζ of $X_S \rightarrow \mathbb{P}^1$. The map $G^c \rightarrow X_\zeta$ given by $g \mapsto \text{Ad}(g)(\Phi(\zeta))$ has differential at the identity $\mathfrak{g}^c \rightarrow T_{\Phi(\zeta)} X_\zeta$ given by $x \mapsto [x, \Phi(\zeta)]$, whose kernel is the centraliser $Z(\Phi(\zeta))$. Note that the regularity condition on Φ ensures the dimension of this space is the same for all $\zeta \in \mathbb{P}^1$. Furthermore, we get a bundle map $\underline{\mathfrak{g}}^c \rightarrow T_F|_\Phi$; here $T_F|_\Phi$ is the restriction to the section Φ of the tangent bundle along the fibres of $X_S \rightarrow \mathbb{P}^1$, and $\underline{\mathfrak{g}}^c$ denotes the trivial bundle of Lie algebras $\mathfrak{g}^c \times \mathbb{P}^1 \rightarrow \mathbb{P}^1$. If we now observe that $T_F|_\Phi$ may be naturally identified with the normal bundle N of the section Φ , we obtain the short exact sequence of holomorphic vector bundles over $\mathbb{P}^1$

$$0 \rightarrow V \rightarrow \underline{\mathfrak{g}}^c \rightarrow N \rightarrow 0, \quad (5)$$

where V is the bundle of centralisers of Φ .

PROPOSITION 5. *The bundle of centralisers is of the form*

$$V \cong \mathcal{O}(-2d_1 + 2) \oplus \mathcal{O}(-2d_2 + 2) \oplus \cdots \oplus \mathcal{O}(-2d_r + 2),$$

where $d_1, d_2, \dots, d_r$ are the exponents of the semisimple Lie algebra $\mathfrak{g}^c$.

Before proving this result let us recall the facts we need about exponents of complex semisimple Lie algebras and invariant polynomials with respect to the adjoint action of G^c ; see [13] and [7].

- Any homogeneous invariant polynomial p , of degree d , can be written in the form $p(X) = f(X, \dots, X)$ ($X \in \mathfrak{g}^c$), where f is an invariant d -linear form.
- The ring of invariant polynomials on $\mathfrak{g}^c$ is freely generated by homogeneous polynomials p_i , for $i = 1, 2, \dots, r$, where $r = \text{rank}(G^c)$, of degrees d_i - the so-called *exponents* of $\mathfrak{g}^c$.
- The exponents d_i satisfy $\sum_{i=1}^r (2d_i - 1) = \dim \mathfrak{g}^c$.

Now for a fixed $Y \in \mathfrak{g}^c$, consider the linear functional $f(Y, \dots, Y, X)$. It can be written as $K(C_Y, X)$, for a unique $C_Y \in \mathfrak{g}^c$. We now observe that the element C_Y belongs to the centraliser of Y . This follows from the calculation $K([C_Y, Y], X) = K(C_Y, [Y, X]) = f(Y, \dots, Y, [Y, X]) = 0$, where we have used the invariance of K and f . We deduce from the non-degeneracy of the Killing form that $[C_Y, Y] = 0$.

LEMMA 6. *If $Y \in \mathfrak{g}^c$ is regular, the r elements C_Y^i , corresponding to the polynomials p_i as before, form a basis for the centraliser of Y in $\mathfrak{g}^c$.*

Proof. Since the centraliser of Y is r -dimensional, we only have to show that the elements C_Y^i ($i = 1, 2, \dots, r$) are linearly independent. This is equivalent to showing the linear independence of the functionals $K(C_Y^i, -) = f_i(Y, \dots, Y, -)$. Notice, however, that in the notation above, $f_i(Y, \dots, Y, -) = (1/d_i)d_Y p_i$. The result now follows from [14], where it is shown that the derivatives $d_Y p_1, \dots, d_Y p_r$ are linearly independent whenever Y is regular. $\square$

Proof of Proposition 5. The section $\Phi : \mathbb{P}^1 \rightarrow \mathfrak{g}^c \otimes \mathcal{O}(2)$ gives rise to a bundle map $\mathcal{O}(2)^* \rightarrow \underline{\mathfrak{g}}^c$, which we still denote by Φ . Since the section Φ is regular, in

particular it is nowhere vanishing, from which we see that the bundle map Φ is an injection. The image of this map, which is a copy of $\mathcal{O}(-2)$, is clearly contained in the bundle of centralisers V . Similarly, for each $i = 1, \dots, r$, $C_{\Phi(\zeta)}^i$ defines an injective bundle map $C_{\Phi}^i : \mathcal{O}(-2)^{\otimes (d_i-1)} \longrightarrow \underline{\mathfrak{g}}^c$, whose image is a subbundle of V isomorphic to $\mathcal{O}(-2(d_i - 1))$. It follows from the Lemma 6 that each fibre V_ζ is generated by $\{C_{\Phi(\zeta)}^i\}_{i=1}^r$, and so we conclude that $V \cong \mathcal{O}(-2(d_1 - 1)) \oplus l \oplus \mathcal{O}(-2(d_r - 1))$. $\square$

Proof of Theorem 4. We shall make use of the exact sequence (5) in order to describe the condition for $\Phi \in H^0(\mathbb{P}^1, \underline{\mathfrak{g}}^c \otimes \mathcal{O}(2))$ to have normal bundle N isomorphic to $\mathcal{O}(1) \oplus l \oplus \mathcal{O}(1)$. We first observe that this is equivalent to the conditions $\deg(N \otimes \mathcal{O}(-1)) = 0$, and $H^0(N \otimes \mathcal{O}(-2)) = 0$; this follows easily from the Grothendieck classification of vector bundles on the projective line. The condition $\deg(N(-1)) = 0$ holds automatically for a regular section Φ . In fact, we see by tensoring the exact sequence (5) by $\mathcal{O}(-1)$ that $\deg N(-1) = \deg \underline{\mathfrak{g}}^c(-1) - \deg V(-1)$. It now follows from Proposition 5 that $\deg V(-1) = \sum_{i=1}^r (-2d_i + 1) = -\dim \underline{\mathfrak{g}}^c = \deg \underline{\mathfrak{g}}^c(-1)$, so that $\deg N(-1) = 0$, as required.

Let us investigate the condition $H^0(N(-2)) = 0$. Tensoring (5) by $\mathcal{O}(-2)$, we get the short exact sequence $0 \longrightarrow V(-2) \longrightarrow \underline{\mathfrak{g}}^c(-2) \longrightarrow N(-2) \longrightarrow 0$. Consider now the sheaf cohomology exact sequence associated to the sequence above. We have in particular

$$H^0(\underline{\mathfrak{g}}^c(-2)) \rightarrow H^0(N(-2)) \rightarrow H^1(V(-2)) \xrightarrow{\alpha} H^1(\underline{\mathfrak{g}}^c(-2)).$$

Since $H^0(\underline{\mathfrak{g}}^c(-2)) = 0$, it follows that $H^1(V(-2)) \xrightarrow{\alpha} H^1(\underline{\mathfrak{g}}^c(-2))$ is injective if and only if $H^0(N(-2)) = 0$. Dualising, we obtain that $H^0(N(-2)) = 0$ if and only if

$$H^0((\underline{\mathfrak{g}}^c)^*) \xrightarrow{\alpha^*} H^0(V^*) \quad (6)$$

is surjective. Since the map α is given by an inclusion (as $V(-2)$ is a subbundle of $\underline{\mathfrak{g}}^c(-2)$), the map α^* is given by restriction. We may canonically identify $H^0((\underline{\mathfrak{g}}^c)^*) = (\underline{\mathfrak{g}}^c)^*$; the map α^* then takes a functional $f \in (\underline{\mathfrak{g}}^c)^*$ to the section ψ of V^* given by $\psi(\zeta) = f|_{V_\zeta}$. Now we easily compute, in view of Proposition 5, that $\dim H^0(V^*) = \sum_{i=1}^r (2d_i - 1) = \dim \underline{\mathfrak{g}}^c$. It thus follows that the map (6) is surjective if and only if it is injective. It *fails* to be injective when there exists a non-zero linear functional $f \in (\underline{\mathfrak{g}}^c)^*$ such that $f(V_\zeta) = 0$ for all $\zeta \in \mathbb{P}^1$. This proves Theorem 4. $\square$

Note that as V_ζ is generated by $\{C_{\Phi(\zeta)}^i\}_{i=1}^r$, the map (6) fails to be injective when $f(C_{\Phi(\zeta)}^i) = 0$ for all $\zeta \in \mathbb{P}^1$, $i = 1, 2, \dots, r = \text{rk}(G^c)$. Therefore, we may reformulate Theorem 4 in a more explicit way as follows.

THEOREM 7. *A real section Φ of X_S is a twistor line if and only if the linear system $K(B, C_{\Phi(\zeta)}^i) = 0$, ($i = 1, 2, \dots, r = \text{rk}(G^c)$, $B \in \mathfrak{g}^c$) only admits the trivial solution $B = 0$.*

In particular, any small deformation of such a section Φ also has normal bundle $\mathcal{O}(1) \oplus \dots \oplus \mathcal{O}(1)$. This is, of course, well known in the context of Kodaira's relative deformation theory. Also note that since C_Y^i is a polynomial in Y of degree $d_i - 1$, and $\Phi(\zeta)$ has degree 2 in ζ , it follows that $C_{\Phi(\zeta)}^i$ has degree $2(d_i - 1)$ in ζ . Therefore, $K(B, C_{\Phi(\zeta)}^i) = 0$ gives us $2(d_i - 1) + 1 = 2d_i - 1$ equations, and hence the total number of equations is $\sum_{i=1}^r (2d_i - 1) = \dim \mathfrak{g}^c$, which is also the number of unknowns of the system. Thus, showing that Φ is a twistor line amounts to checking the non-vanishing of a determinant.

EXAMPLE. Let $\mathfrak{g}^c = \mathfrak{sl}(k, \mathbb{C})$. Then the condition for Φ to be a twistor line reads: If $B \in \mathfrak{sl}(k, \mathbb{C})$ is such that $\text{Tr}(B\Phi^l) = 0$ for all $\zeta \in \mathbb{P}^1$ and $l = 1, 2, \dots, k - 1$, then $B = 0$.

5. The Metric

The starting point of the link between hyperkähler geometry and twistor geometry is, as discussed in [9], the identification of a hyperkähler manifold M with the parameter space of real twistor lines of its twistor space Z (at least locally). The complexified tangent space to M , at a point which is represented in the correspondence above by the twistor line Φ , is identified with the space $H^0(N)$ of sections of the normal bundle of Φ . (We recall that we may identify $N \cong T_F|_{\Phi}$, where T_F is the tangent bundle along the fibres of $Z \rightarrow \mathbb{P}^1$; thus, roughly speaking, a section of the normal bundle is an infinitesimal change of Φ in the direction of the fibres of Z .)

In order to discuss the twistor geometry of X_S , we first describe the space $H^0(N)$. Since $N \cong N(-1) \otimes \mathcal{O}(1)$, where $N(-1)$ is trivial, we have a canonical decomposition $H^0(N) \cong H^0(N(-1)) \otimes H^0(\mathcal{O}(1))$. It will be useful to describe the global sections of $N(-1)$ in terms of the bundle of centralisers V . Note that tensoring (5) by $\mathcal{O}(-1)$, we get the exact sequence $0 \rightarrow V(-1) \rightarrow \underline{\mathfrak{g}}^c(-1) \rightarrow N(-1) \rightarrow 0$.

LEMMA 8. *There is a correspondence between sections $a \in H^0(N(-1))$ and pairs (a_+, a_-) , where $a_+ : U_0 \rightarrow \mathfrak{g}^c$, $a_- : U_\infty \rightarrow \mathfrak{g}^c$ are such that $a_+(\zeta) - \zeta^{-1}a_-(\zeta^{-1}) \in V_\zeta$ on $U_0 \cap U_\infty$. Here, a_+ and a_- are unique up to the addition of local sections of V , defined on U_0 and U_∞ , respectively.*

This is elementary, but for lack of a reference we shall supply a proof.

SUBLEMMA 9. *Let W be a vector bundle on $\mathbb{P}^1$. There is a 1–1 correspondence between sections $s \in H^0(\mathbb{P}^1; W(-1))$ and pairs (ϕ_+, ϕ_-) of local holomorphic*

sections $\phi_+ : U_0 \rightarrow W$, $\phi_- : U_\infty \rightarrow W$ such that $\phi_+(\zeta) = \zeta^{-1}\phi_-(\zeta^{-1})$ on $U_0 \cap U_\infty$.

Proof. A section $s \in H^0(\mathbb{P}^1; W(-1))$ may be interpreted as a bundle map $s : \mathcal{O}(1) \rightarrow W$. Describing $\mathcal{O}(1)$ by means of the usual coordinate patches, the result follows easily. $\square$

Proof of Lemma 8. If (a_+, a_-) are as in the lemma, then passing to the quotient by V , we get maps $\bar{a}_+ : U_0 \rightarrow \mathfrak{g}^c/V$, $\bar{a}_- : U_\infty \rightarrow \mathfrak{g}^c/V$ satisfying $\bar{a}_+(\zeta) = \zeta^{-1}\bar{a}_-(\zeta^{-1})$. Applying the sublemma above with $W = \mathfrak{g}^c/V = N$, we see that the pair $(\bar{a}_+, \bar{a}_-)$ defines a section of $N(-1)$. Conversely, if $s \in H^0(N(-1))$, the sublemma gives us a pair $\phi_+ : U_0 \rightarrow \mathfrak{g}^c/V$, $\phi_- : U_\infty \rightarrow \mathfrak{g}^c/V$ such that $\phi_+(\zeta) = \zeta^{-1}\phi_-(\zeta^{-1})$. Now since the subbundle V of $\underline{\mathfrak{g}}^c$ may be holomorphically trivialised over U_0 , we may find a complementary subbundle W to V in $\underline{\mathfrak{g}}^c$ over U_0 , so that $\mathfrak{g}^c/V \cong W$ on U_0 and we may write $a_+ : U_0 \rightarrow W \subset \underline{\mathfrak{g}}^c$. Similarly we may represent a_- as a map $a_- : U_\infty \rightarrow \underline{\mathfrak{g}}^c$. It is clear that $\phi_+ = \bar{a}_+$, $\phi_- = \bar{a}_-$. Finally, the uniqueness statement of the lemma is immediate. $\square$

Let us now describe global sections of N , using Lemma 8 above. Write $s \in H^0(N)$ as $s = a \otimes 1 + b \otimes \zeta$, where $a, b \in H^0(N(-1))$, and let a, b be described by the pairs (a_+, a_-) and (b_+, b_-) respectively, as before. We may represent s by $C_+(\zeta) = a_+ + b_+\zeta$ on U_0 , $-C_-(\zeta) = a_-\zeta + b_-$ on U_∞ . On $U_0 \cap U_\infty$, we have $C_+(\zeta) + C_-(\zeta^{-1}) = a_+(\zeta) - \zeta^{-1}a_-(\zeta^{-1}) + \zeta(b_+(\zeta) - b_-(\zeta^{-1})\zeta^{-1})$, so that

$$C_+(\zeta) + C_-(\zeta^{-1}) \in V_\zeta \text{ on } U_0 \cap U_\infty. \quad (7)$$

By abuse of notation, we shall call (C_+, C_-) the pair associated to s ; in fact, such a pair is defined by the choice of the pairs (a_+, a_-) , (b_+, b_-) , and the latter are unique up to addition of local sections of V , as observed earlier. The formulas we shall work with below are independent of such choices. Once one has identified the complexified tangent space with the space of sections of the normal bundle of the twistor line, one proceeds to define the metric in two steps (as described in [9]):

- First we construct a complex inner product g on $H^0(N)$. The twistorial data necessary to achieve this is the so-called twisted two-form along the fibres of $Z \rightarrow \mathbb{P}^1$, $\omega \in H^0(Z; \wedge^2 T_F^*(2))$. By means of the identification $N \cong T_F$, we obtain a complex-symplectic form on $H^0(\Phi, N(-1))$, which we still denote by ω ; this together with the natural symplectic structure on $H^0(\mathcal{O}(1))$ defines, in a canonical fashion, a complex inner product on $T_m M \otimes \mathbb{C} = H^0(N(-1)) \otimes H^0(\mathcal{O}(1))$, given by the formula $g(a + b\zeta, a + b\zeta) = 2\omega(a, b)$, where $\{1, \zeta\}$ is the canonical basis of $H^0(\mathcal{O}(1))$ and $a, b \in H^0(N(-1))$.
- Next we use a quaternionic structure j on $H^0(N(-1))$ and the usual quaternionic structure on $H^0(\mathcal{O}(1))$ to obtain a real structure t on $T_m M \otimes \mathbb{C} = H^0(N(-1)) \otimes H^0(\mathcal{O}(1))$, given by $t(a + b\zeta) = j(b) - j(a)\zeta$, so that a real tangent vector may be written as $X = a - j(a)\zeta$, $a \in H^0(\Phi; N(-1))$, and the metric is given by $g(X, X) = -2\omega(a, j(a))$. One has to check that g is

real-valued and positive-definite; this is what in twistor language is referred to as the compatibility of the twisted two-form ω with the real structure.

5.1. THE TWISTED 2-FORM

Let us be given two sections $s, t \in H^0(N)$, represented by the pairs (C_+, C_-) and (D_+, D_-) , as before. We define the twisted 2-form ω by

$$\omega(s, t) = K([C_+, D_+], \Phi). \quad (8)$$

This is the required twisted holomorphic symplectic form. If we consider its restriction to a fibre $X_\zeta = \mathcal{O}_{\Phi(\zeta)}$, this is simply the Kostant–Kirillov form (hence is closed and non-degenerate, see [10]). However, we must check that globally the expression (8) defines an $\mathcal{O}(2)$ -valued form.

LEMMA 10. *The expression (8) defines a 2-form ω with values in $\mathcal{O}(2)$.*

Proof. We want to show that the expression $K([C_+, D_+], \Phi)$ is a quadratic polynomial in ζ . It will suffice to show that

$$K([C_+(\zeta), D_+(\zeta)], \Phi(\zeta)) = K([C_-(\zeta^{-1}), D_-(\zeta^{-1})], \Phi(\zeta)),$$

since the left-hand side is a polynomial in ζ , and the right-hand side does not contain terms of degree > 2 in ζ . However, this is clear in view of (7) and the invariance of K . $\square$

The induced complex symplectic form ω on $H^0(N(-1))$ is described similarly, as follows. If a and b are two sections of $N(-1)$, represented by the pairs (a_+, a_-) and (b_+, b_-) , respectively, then $\omega(a, b) = K([a_+, b_+], \Phi) \in \mathbb{C}$. In fact, one may argue as in the proof of the lemma above to show that

$$\omega(a, b) = K([a_-, b_-], \tilde{\Phi}) = \frac{1}{\zeta^2} K([a_-, b_-], \Phi), \quad (9)$$

from which we see that $\omega(a, b) \in \mathbb{C}$.

5.2. THE QUATERNIONIC STRUCTURE

Let $a \in H^0(N(-1))$ be represented by $a_+ : U_0 \rightarrow \mathfrak{g}^c$, $a_- : U_\infty \rightarrow \mathfrak{g}^c$, satisfying $a_+(\zeta) - \zeta^{-1}a_-(\zeta^{-1}) \in V_\zeta$. Let us, for convenience, expand these maps in power series $a_+ = \sum_{i \geq 0} a_{+,i} \zeta^i$, $a_- = \sum_{i \geq 0} a_{-,i} \tilde{\zeta}^i$, ($a_{+,i}, a_{-,i} \in \mathfrak{g}^c$). Now define

$$j(a)_+ = a_-^*(-\bar{\zeta}) = \sum_{i \geq 0} a_{-,i}^*(-\zeta)^i, \quad j(a)_- = -a_+^*(-\bar{\zeta}) = -\sum_{i \geq 0} a_{+,i}^*(-\tilde{\zeta})^i.$$

LEMMA 11. *The pair $(j(a)_+, j(a)_-)$ as above defines a section $j(a)$ of $N(-1)$. The map $j : H^0(N(-1)) \rightarrow H^0(N(-1))$ is a quaternionic structure.*

Proof. We have to check that $j(a)_+(\zeta) - \zeta^{-1}j(a)_-(\zeta^{-1}) \in V_\zeta$. Now since $[a_+(\zeta) - \zeta^{-1}a_-(\zeta^{-1}), \Phi(\zeta)] = 0$, we get by multiplying by ζ and taking $*$, $[(a_-(\zeta^{-1}))^* - \bar{\zeta}(a_+(\zeta))^*, (\Phi(\zeta))^*] = 0$. If we now evaluate the expression on the left at $-\bar{\zeta}^{-1}$, and use the fact that Φ satisfies the reality condition $\Phi(-\bar{\zeta}^{-1})^* = -(\Phi(\zeta)/\zeta^2)$, we obtain $[j(a)_+(\zeta) - \zeta^{-1}j(a)_-(\zeta^{-1}), \Phi(\zeta)] = 0$, as required. It is clear that j defines a quaternionic structure on $H^0(N(-1))$, that is, j is real-linear, $j^2 = -1$ and $j(\lambda a) = \bar{\lambda}j(a)$ for $\lambda \in \mathbb{C}$. $\square$

5.3. THE COMPLEX INNER PRODUCT AND THE METRIC

We write $u \in H^0(N)$ as $u = a \otimes 1 + b \otimes \zeta$, for $a, b \in H^0(N(-1))$. The complex inner product g is given by $g(u, u) = 2\omega(a, b) = 2K([a_+, b_+], \Phi) \in \mathbb{C}$.

The metric is given, for a *real* tangent vector $u = a - j(a)\zeta$, where $a \in H^0(N(-1))$ is represented by the pair (a_+, a_-) , by $g(u, u) = -2\omega(a, j(a)) = -2K([a_+, j(a)_+], \Phi) = -2K([a_+(\zeta), a_-^*(-\bar{\zeta})], \Phi(\zeta))$. If we write $\Phi = X_0 + X_1\zeta + X_2\zeta^2$, we obtain

$$g(u, u) = -2K([a_{+,0}, a_{-,0}^*], X_0). \quad (10)$$

We have used the fact that the terms in ζ^i , for $i \neq 0$, must vanish in the expression $-2K([a_+(\zeta), a_-^*(-\bar{\zeta})], \Phi(\zeta))$; see (9).

LEMMA 12. *The restriction of g to real tangent vectors is real-valued.*

Proof. We have $K([a_+, j(a)_+], \Phi) = K([a_{+,0}, a_{-,0}^*], X_0)$. We must show this expression equals its complex conjugate. Using (9), we get $K([a_+, j(a)_+], \Phi) = (1/\zeta^2)K([a_-, j(a)_-], \Phi) = K([a_{-,0}, -a_{+,0}^*], X_2)$. Since $X_2 = -X_0^*$ (as $\Phi = X_0 + X_1\zeta + X_2\zeta^2$ is a real section) and the Killing form satisfies $K(X^*, Y^*) = \overline{K(X, Y)}$, our claim is proved. $\square$

We have not, however, addressed the question of positive-definiteness of g ; once this is settled the arguments in [9] show that g is indeed a hyperkähler metric. We shall prove positivity in the case $G^c = SL(2, \mathbb{C})$, in the Example below. For the general case, see the comments at the end of this section.

Before that, let us see how to write down the expression for g in a more explicit way. We restrict ourselves, for simplicity, to the case $G^c = SL(k, \mathbb{C})$, but the discussion may be extended to any semisimple group. Let us be given $a \in H^0(N(-1))$, represented by the pair $a_+ : U_0 \rightarrow \mathfrak{g}^c$ and $a_- : U_\infty \rightarrow \mathfrak{g}^c$. For $G^c = SL(k, \mathbb{C})$, the fibre V_ζ of the bundle of centralisers V is generated by $\Phi(\zeta), \Phi(\zeta)^2 - (1/k)\text{Tr}(\Phi(\zeta)^2)I_k, \dots, \Phi(\zeta)^{k-1} - (1/k)\text{Tr}(\Phi(\zeta)^{k-1})I_k$. As $a_+(\zeta) - \zeta^{-1}a_-(\zeta^{-1}) \in V_\zeta$ we can write

$$a_+(\zeta) - \zeta^{-1}a_-(\zeta^{-1}) = \sum_{i=1}^{k-1} \alpha_i(\zeta, \zeta^{-1})(\Phi(\zeta))^i - \frac{1}{k} \text{Tr}(\Phi(\zeta)^i)I_k).$$

Reciprocally, writing

$$\begin{aligned} \sum_{i=1}^{k-1} \alpha_i(\zeta, \zeta^{-1})(\Phi(\zeta)^i - (1/k)\mathrm{Tr}(\Phi(\zeta)^i)I_k) &= \sum_{-\infty}^{\infty} \beta_i \zeta^i \\ &= \zeta^{-1} \sum_0^{\infty} \beta_{-i-1} \zeta^{-i} + \sum_0^{\infty} \beta_i \zeta^i, \end{aligned}$$

we see that if we take $a_+(\zeta) = \sum_0^{\infty} \beta_i \zeta^i$ and $-a_-(\zeta^{-1}) = \sum_0^{\infty} \beta_{-i-1} \zeta^{-i}$, then (a_+, a_-) defines an $a \in H^0(N(-1))$. Since a_+ and a_- are defined modulo V , it is easy to see that it is enough to consider coefficients α_i which are polynomials in ζ^{-1} of degree not exceeding $2i$. Also note from (10) that for the purpose of computing the metric g , we only need to write down $a_{+,0}$ and $a_{-,0}$.

EXAMPLE. Let us be more explicit in the case $G^c = SL(2, \mathbb{C})$; we shall be able to prove definiteness of g in this case. We have

$$\begin{aligned} a_+(\zeta) - \zeta^{-1}a_-(\zeta^{-1}) &= \alpha(\zeta, \zeta^{-1})\Phi(\zeta) \\ &= \gamma\zeta^{-1}\Phi(\zeta) + \delta\zeta^{-2}\Phi(\zeta) \\ &= \gamma X_0\zeta^{-1} + \gamma X_1 + \gamma X_2\zeta + \delta X_0\zeta^{-2} + \delta X_1\zeta^{-2} + \delta X_2, \end{aligned}$$

for $\gamma, \delta \in \mathbb{C}$. We thus have

$$\begin{aligned} a_+(\zeta) &= (\delta X_1 + \gamma X_2) + \delta X_2\zeta; \\ -a_-(\zeta) &= (\delta X_0 + \gamma X_1) + \gamma X_0\zeta. \end{aligned}$$

We compute

$$\begin{aligned} g(u, u) &= -2\mathrm{Tr}([a_{+,0}, a_{-,0}^*], X_0) \\ &= -2\mathrm{Tr}([\delta X_1 + \gamma X_2, \bar{\delta} X_2 - \bar{\gamma} X_1], X_0) \\ &= -2(|\gamma|^2 + |\delta|^2)\mathrm{Tr}([X_1, X_2], X_0). \end{aligned}$$

We claim that $\mathrm{Tr}([X_1, X_2], X_0)$ *never vanishes on twistor lines*. This will prove the definiteness of g . We prove our claim by contradiction. If $\mathrm{Tr}([X_1, X_2], X_0) = 0$, then since $\Phi = X_0 + X_1\zeta + X_2\zeta^2$, we clearly see that $\mathrm{Tr}([X_1, X_2], \Phi) = 0$. However, as Φ is a twistor line, then as we saw in the Example in Section 4, one must have $[X_1, X_2] = 0$. Now we can also show that $[X_0, X_2] = 0$. Indeed, we have $\mathrm{Tr}([X_0, X_2]\Phi) = \mathrm{Tr}([X_0, X_2]\zeta X_1) = \zeta\mathrm{Tr}([X_2, X_1]X_0) = 0$; hence we may argue as before to obtain $[X_0, X_2] = 0$. Now since by hypothesis X_2 is regular, then since X_0 and X_1 belong to the one-dimensional centraliser of X_2 , they must be of the form $X_i = \mu_i X_2$ for $\mu_i \in \mathbb{C}$, $i = 0, 1$. However, it then follows that $\Phi = (\mu_0 + \mu_1\zeta + \zeta^2)X_2$; but then Φ vanishes at the roots of $\mu_0 + \mu_1\zeta + \zeta^2$,

contradicting its regularity. This proves our claim. We have thus seen that in the case $G^c = SL(2, \mathbb{C})$, g defines a (positive-definite) hyperkähler metric.

For more general groups G^c , however, it may be necessary to require that Φ satisfies further constraints, other than being a real line with the required normal bundle, in order for g to be a (positive-definite) hyperkähler metric on a neighbourhood of Φ . In any case, once we write down a twistor line Φ explicitly, it is easy (although tedious) to check positivity; in fact we see, by performing the sort of calculations done in the Example above, that this amounts to verifying that a certain hermitian matrix, whose coefficients only depend on Φ , is positive. Once this is checked, the results in [9] show that g is a hyperkähler metric. It would be interesting to determine explicitly what those constraints are which ensure positivity of g .

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